# Circuit Diagrams & Circuit Components

**Exam code:** 0625 & 0972

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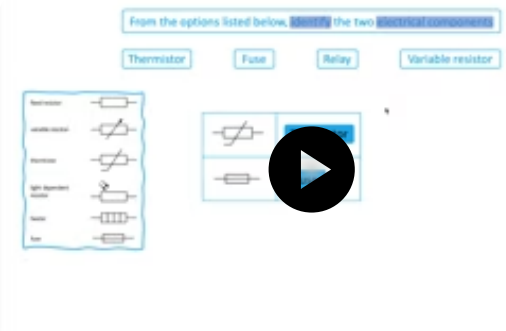


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## Circuit components

### What are circuit symbols?

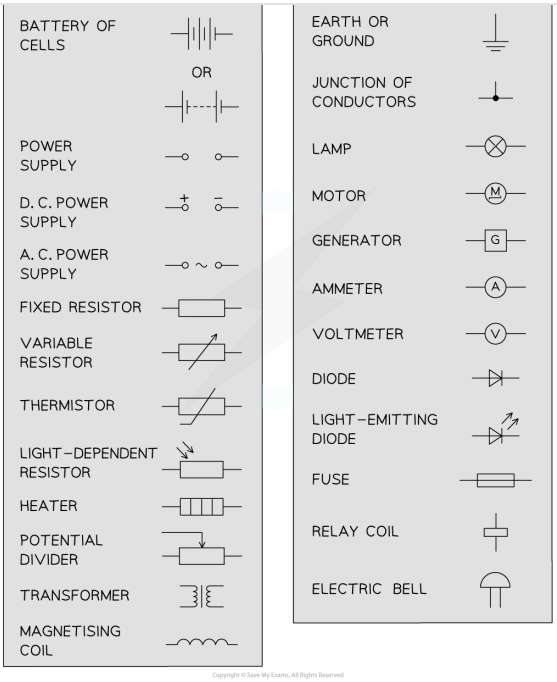
- The diagram below shows the circuit symbols that could be used in circuit diagrams to represent circuit **components**
  - You will be expected to know **what each component is** and **how it behaves** in a circuit

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it




*Circuit symbols represent these components in circuit diagrams*

# Types of circuit components

## Power supplies

- Cells, batteries, power supplies and generators all **supply current** to the circuit

## Resistors

- Potential dividers, fixed and variable resistors, thermistors and light-dependent resistors (LDRs) are all used to **control current**

## Meters

- Ammeters and voltmeters are used to measure the current and potential difference
  - Ammeters are always connected in series whilst voltmeters are always connected in parallel

## Switches and functional components

- Switches open and close a circuit and determine whether current can flow
- Functional components perform specific roles when current passes through them:
  - Motors rotate
  - Lamps emit light
  - Heaters transfer thermal energy
  - Bells emit a sound

## Electromagnetic components

- Magnetising coils, relays and transformers use electromagnetic effects

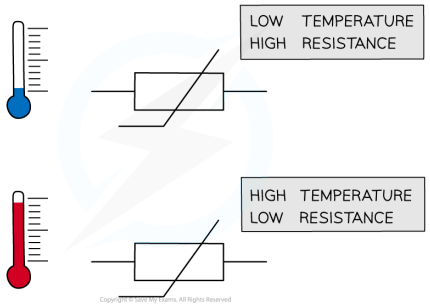
potential difference

## Fuses

- Protect expensive components from current surges and act as a safety measure against fire

## Thermistors

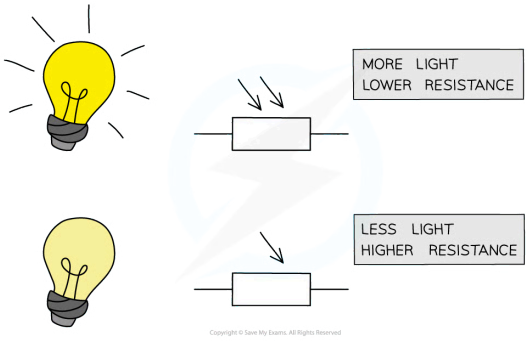
- A thermistor is a **non-ohmic** conductor and a temperature-dependent resistor
- The resistance of a thermistor changes depending on its **temperature**
  - As the temperature **increases** the resistance of a thermistor **decreases** and vice versa




***The resistance through a thermistor is dependent on the temperature of it***

## Light-dependent resistors

- A light-dependent resistor (LDR) is a non-ohmic conductor and sensory resistor
- Its resistance automatically changes depending on the light energy falling onto it (illumination)
- As the **light intensity increases**, the **resistance** of an LDR **decreases**




***Resistance of an LDR depends on the light intensity falling on it***

# Diodes

and draw the circuit symbol for a diode:

***A diode is a component that only allows a current in one direction***

- Diodes are occasionally drawn with a horizontal line running through the middle of them
- Diodes only allow current to flow through them in **one** direction (see the diagram below)
  - This is the direction that the triangle points in the diagram

***When travelling in the same direction as the diode, current can flow. If the diode is flipped, it now prevents current from flowing in the circuit***

- Light emitting diodes (LEDs) behave the exact same way, but they emit light when current flows through them

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